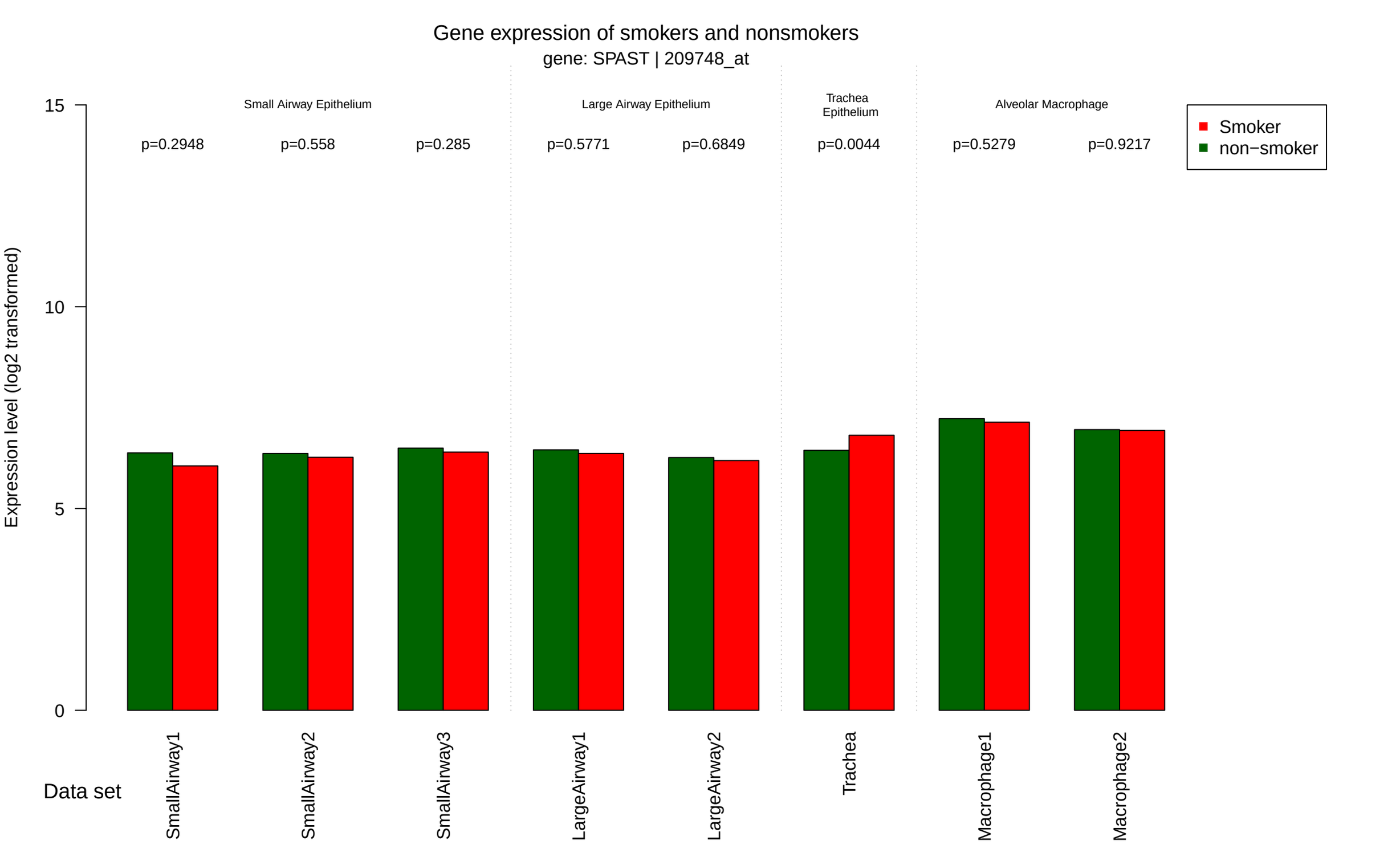
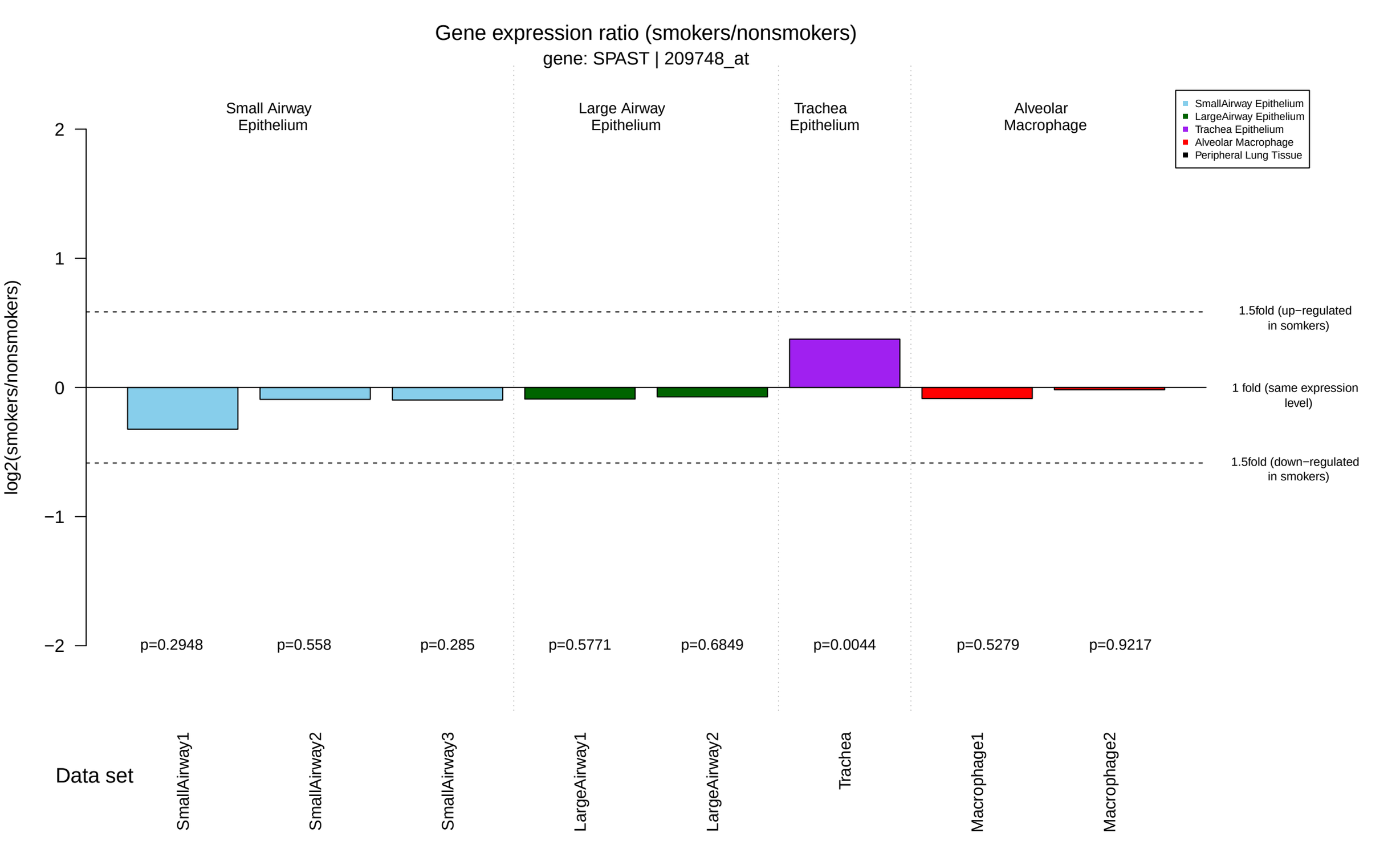
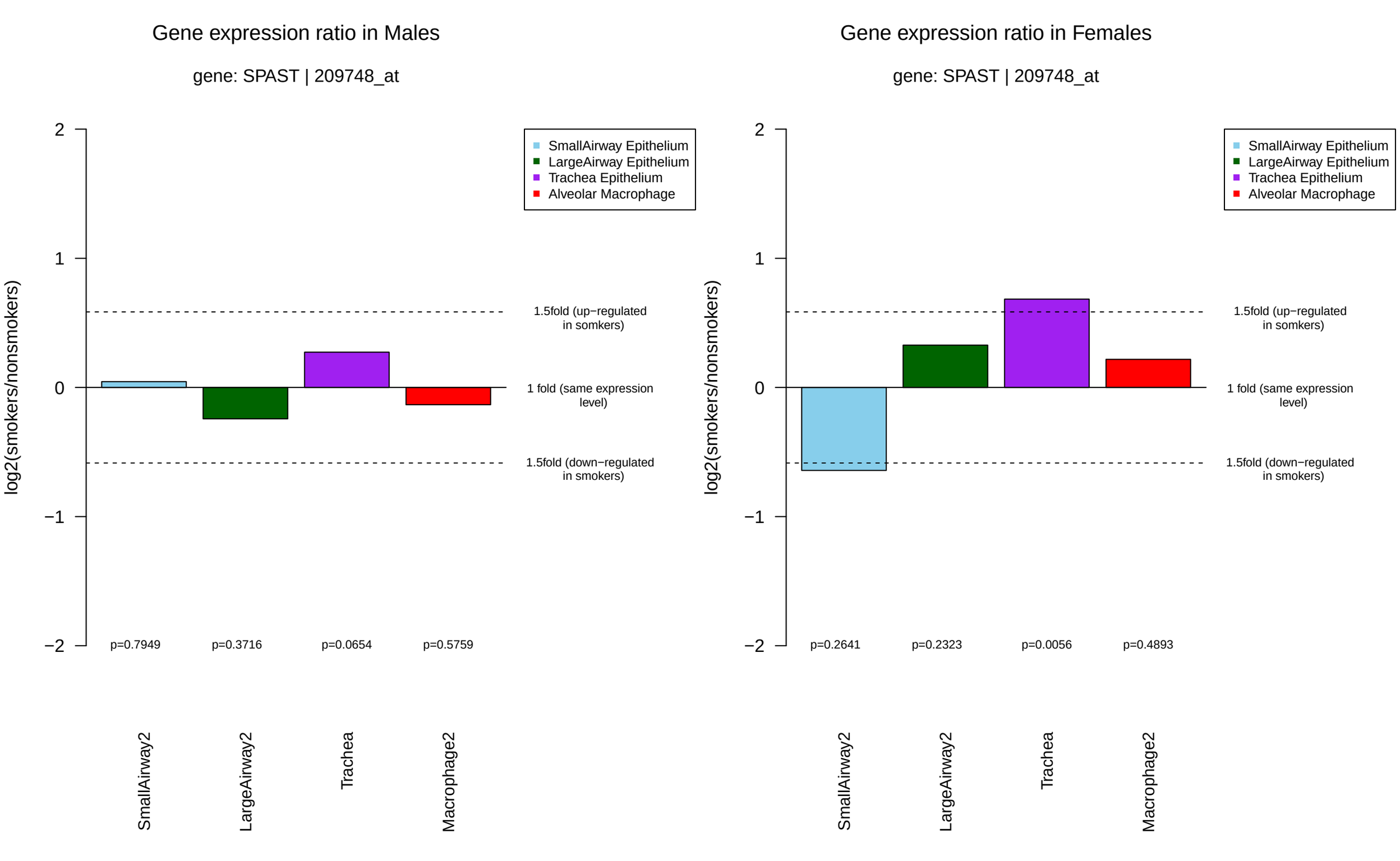
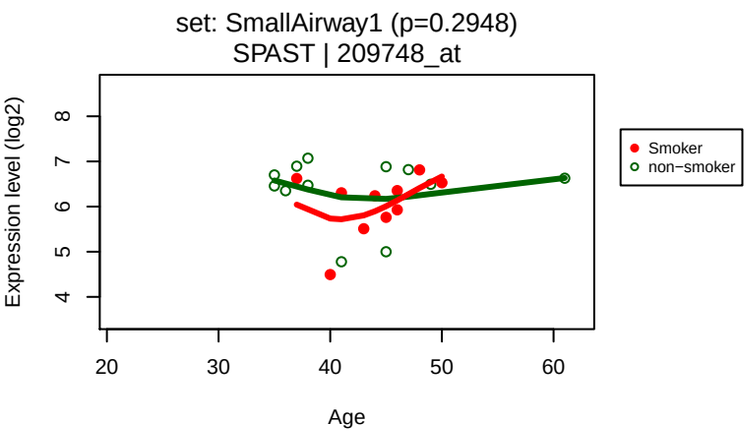
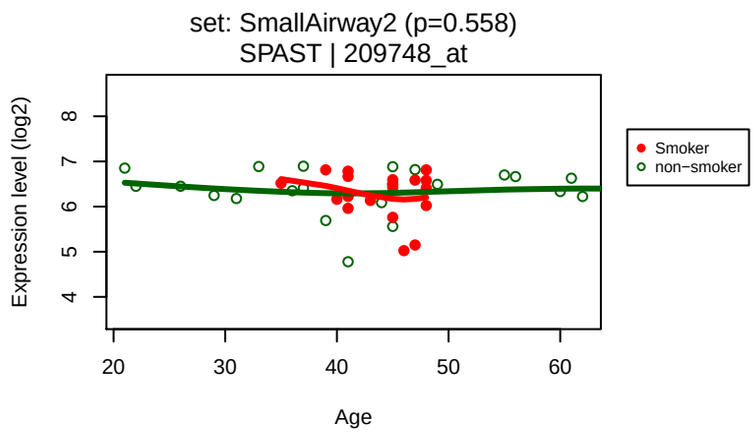
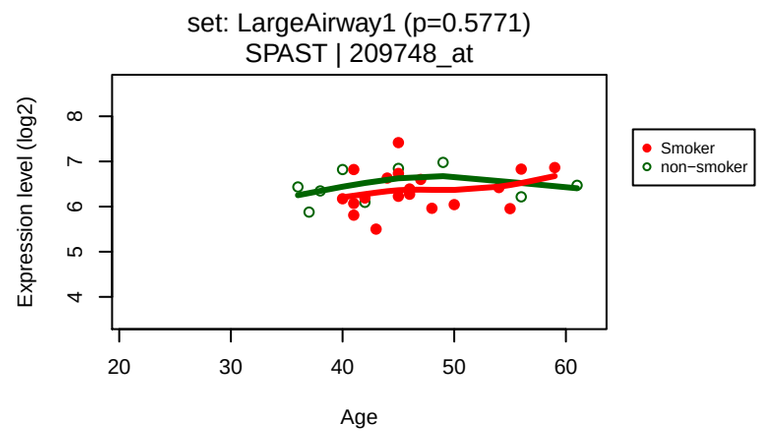

Gene expression of smokers and nonsmokers
gene: SPAST | 209748_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2948
p=0.558
p=0.285
p=0.5771
p=0.6849
p=0.0044
p=0.5279
p=0.9217
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

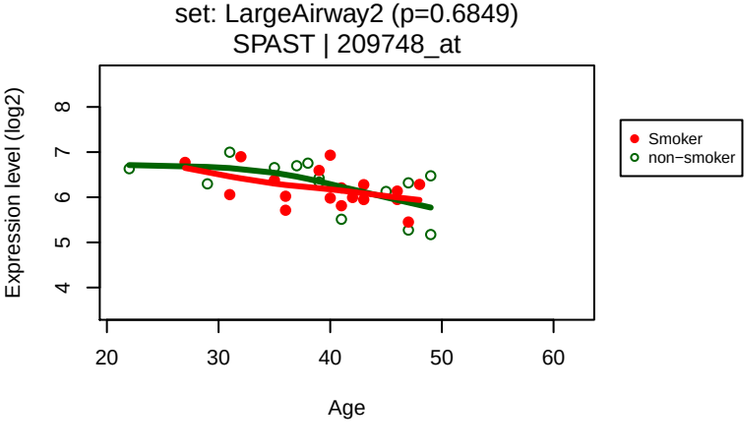

Gene expression ratio (smokers/nonsmokers)
gene: SPAST | 209748_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2948
p=0.558
p=0.285
p=0.5771
p=0.6849
p=0.0044
p=0.5279
p=0.9217
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

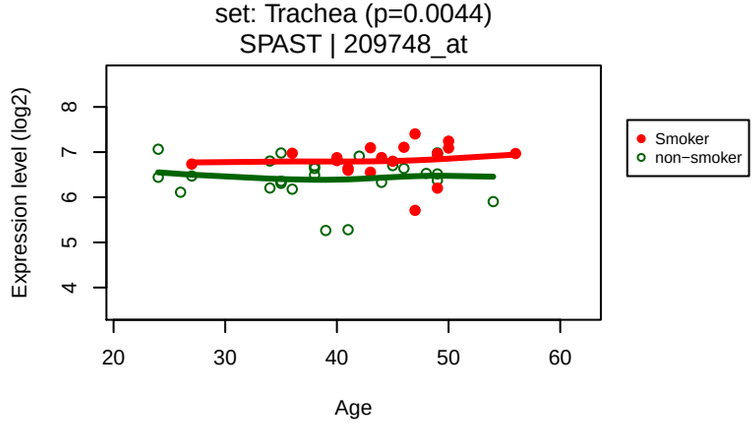

Gene expression ratio in Males
gene: SPAST | 209748_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7949
p=0.3716
p=0.0654
p=0.5759
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPAST | 209748_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2641
p=0.2323
p=0.0056
p=0.4893
SmallAirway2
LargeAirway2
Trachea
Macrophage2

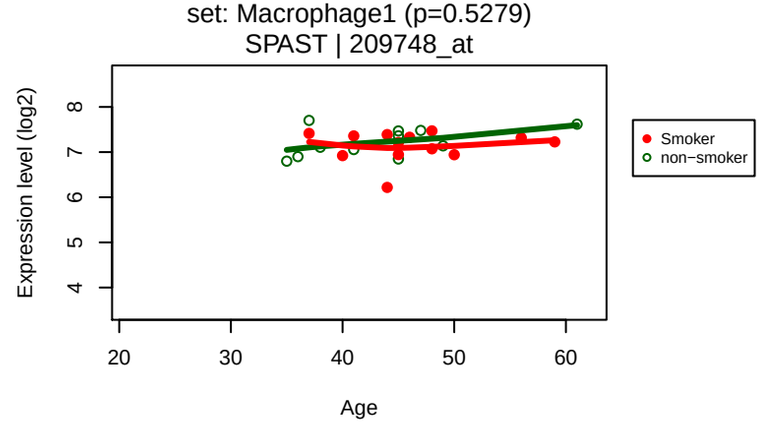

set: SmallAirway1 (p=0.2948)
SPAST | 209748_at
Expression level (log2)
Age
Smoker
non−smoker

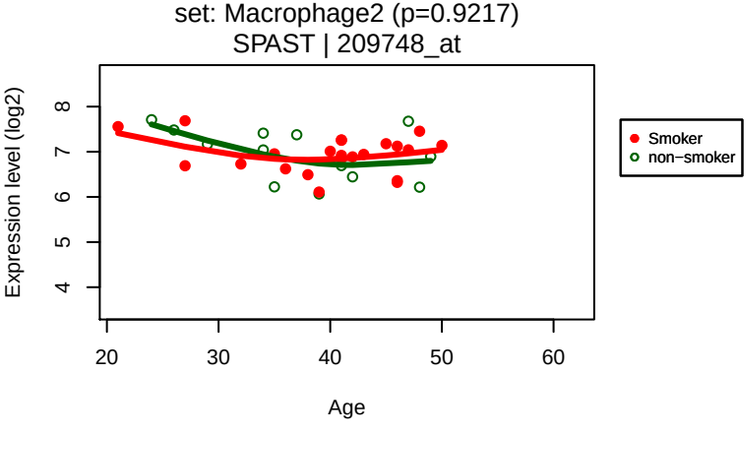

set: SmallAirway2 (p=0.558)
SPAST | 209748_at
Expression level (log2)
Age
Smoker
non−smoker

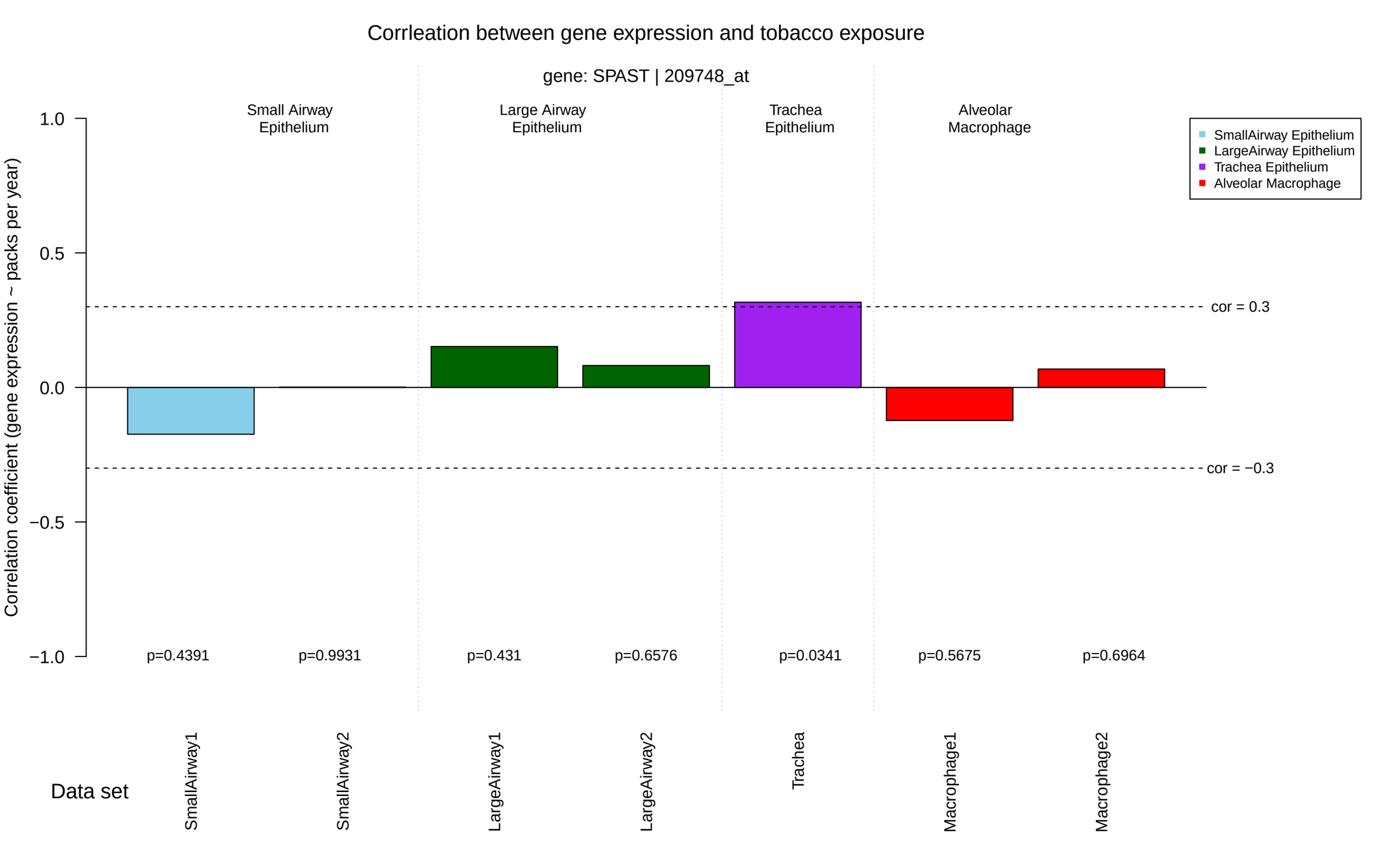

set: LargeAirway1 (p=0.5771)
SPAST | 209748_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6849)
SPAST | 209748_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0044)
SPAST | 209748_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5279)
SPAST | 209748_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9217)
SPAST | 209748_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPAST | 209748_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
Correlation coefficient (gene expression ~ packs per year)
cor = 0.3
cor = −0.3
p=0.4391
p=0.9931
p=0.431
p=0.6576
p=0.0341
p=0.5675
p=0.6964
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2